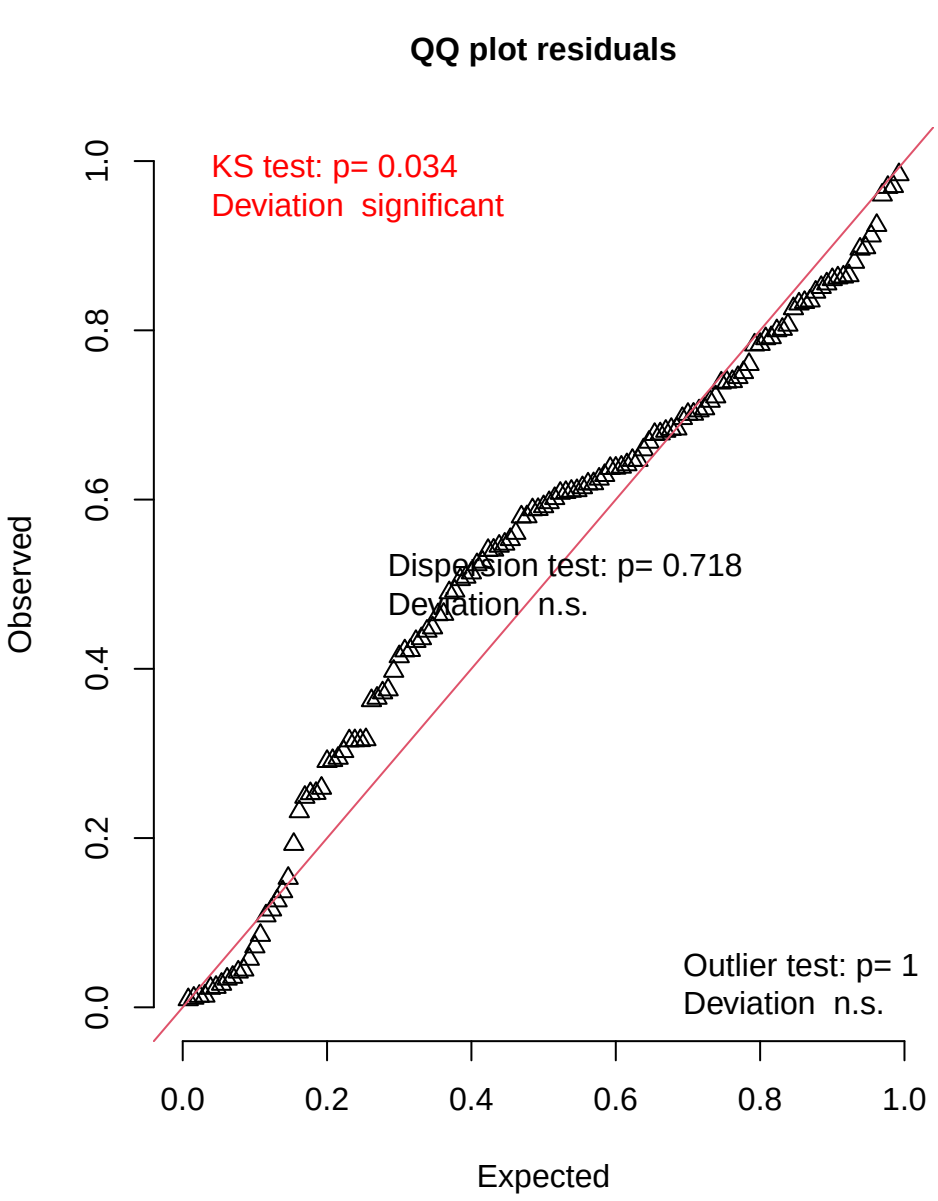
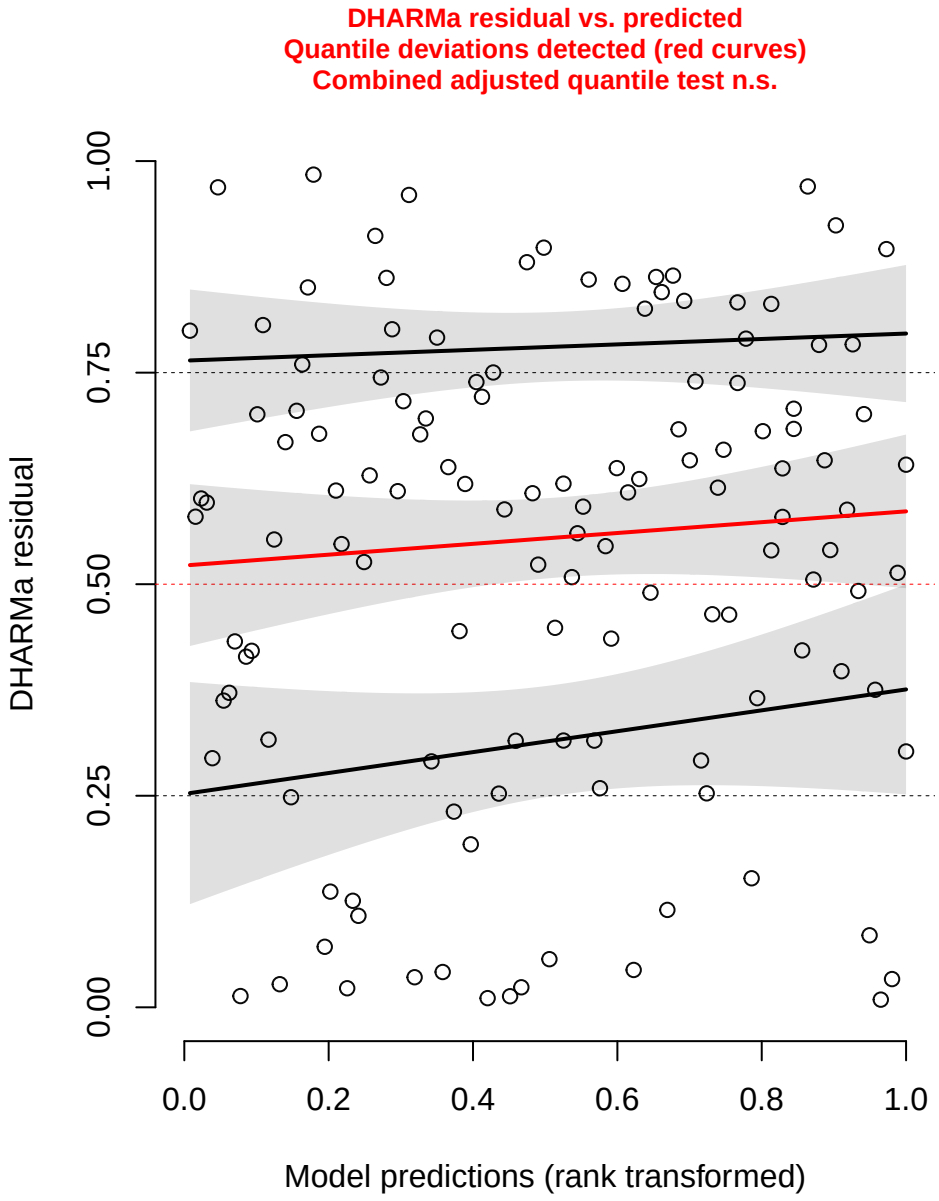
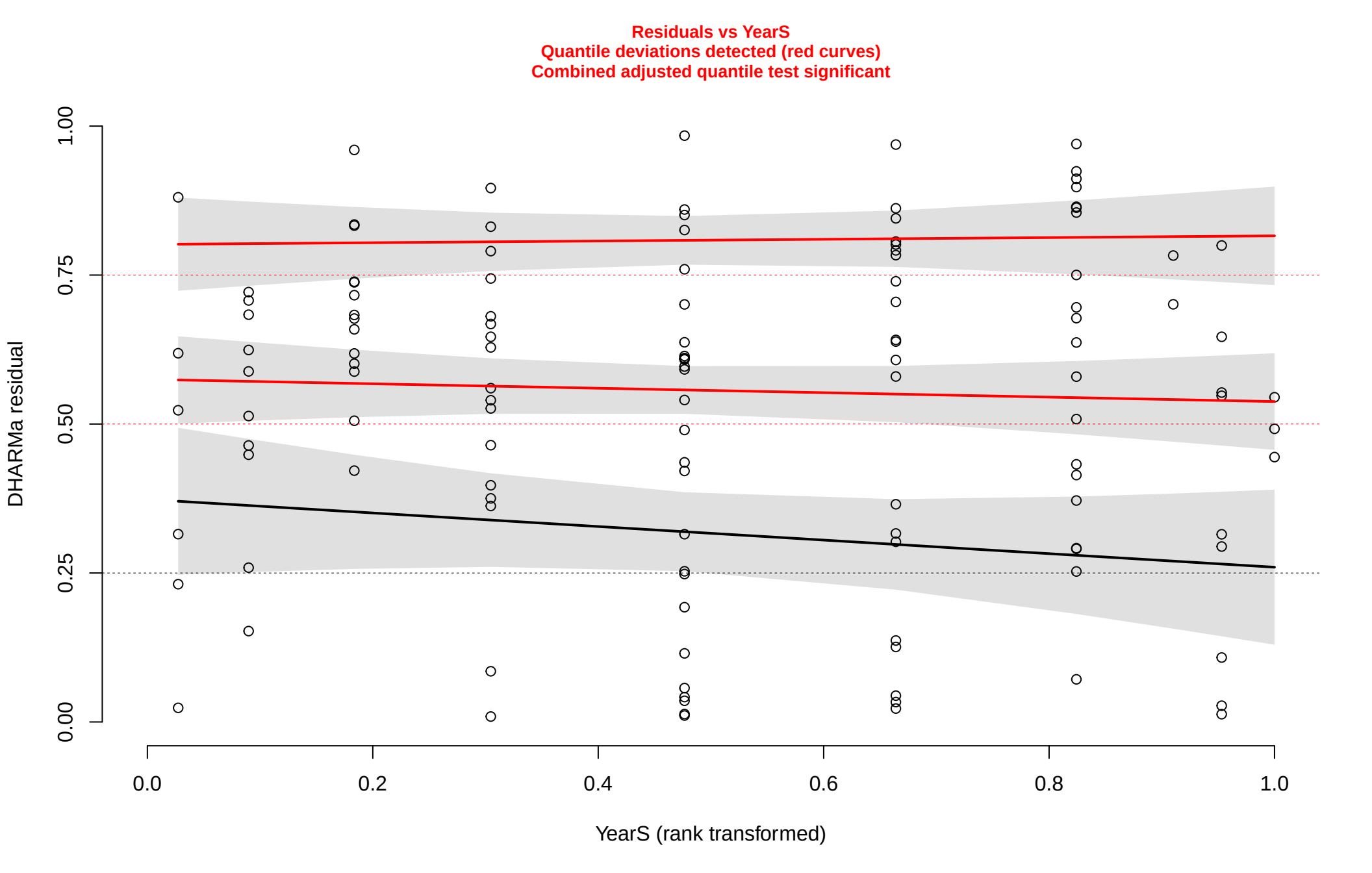
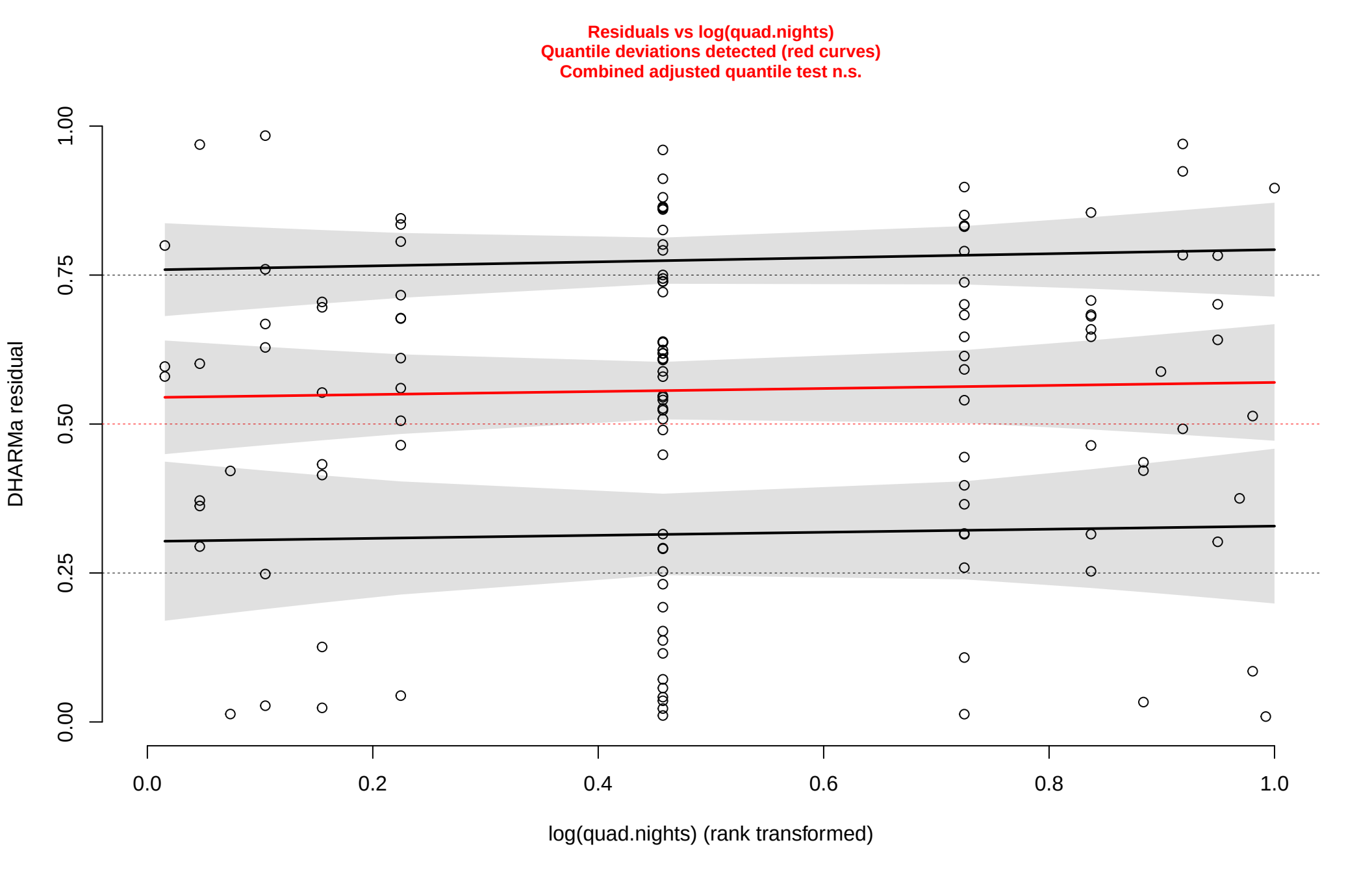
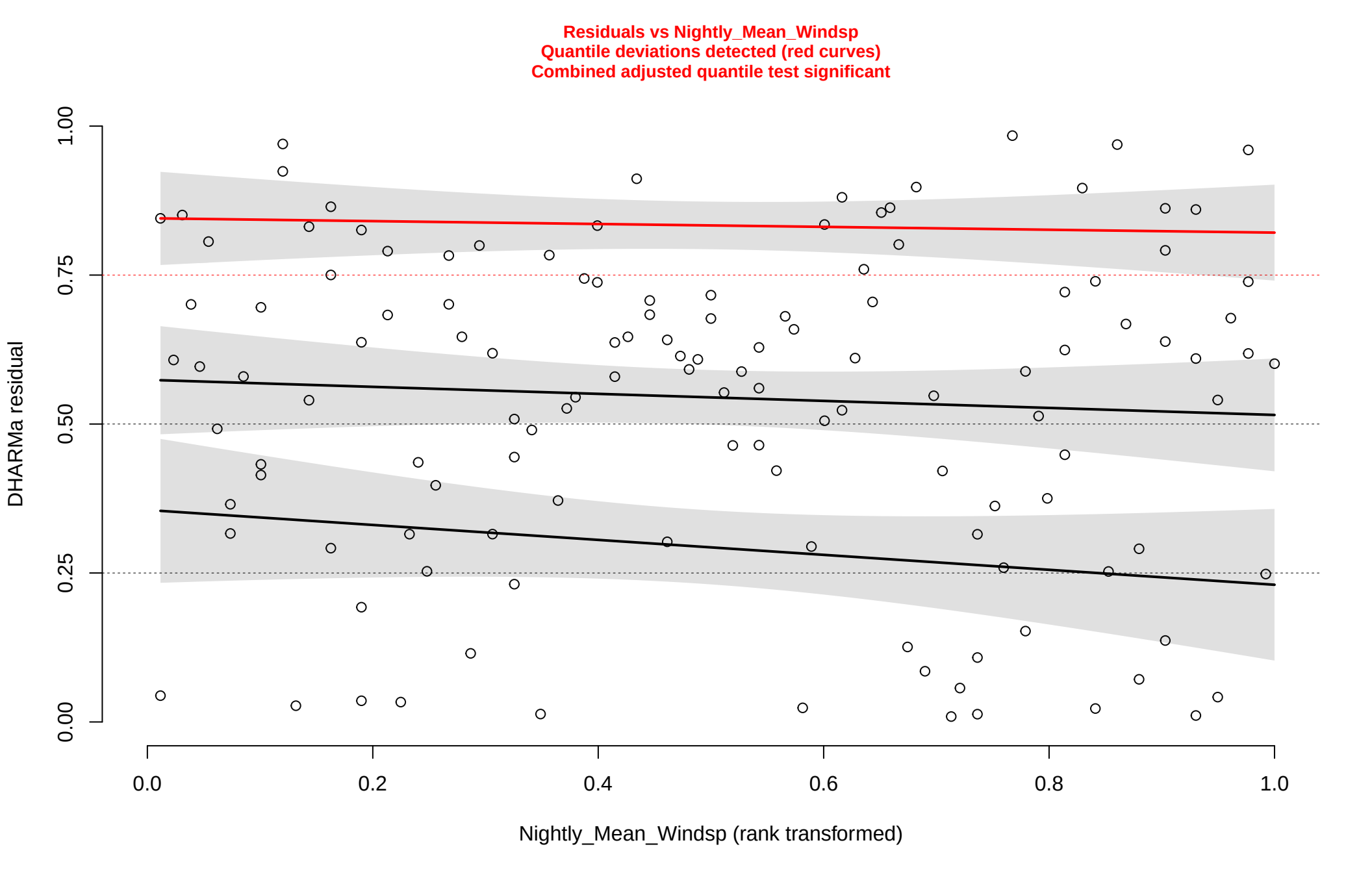
DHARMa residuals: Northern Region_LABO

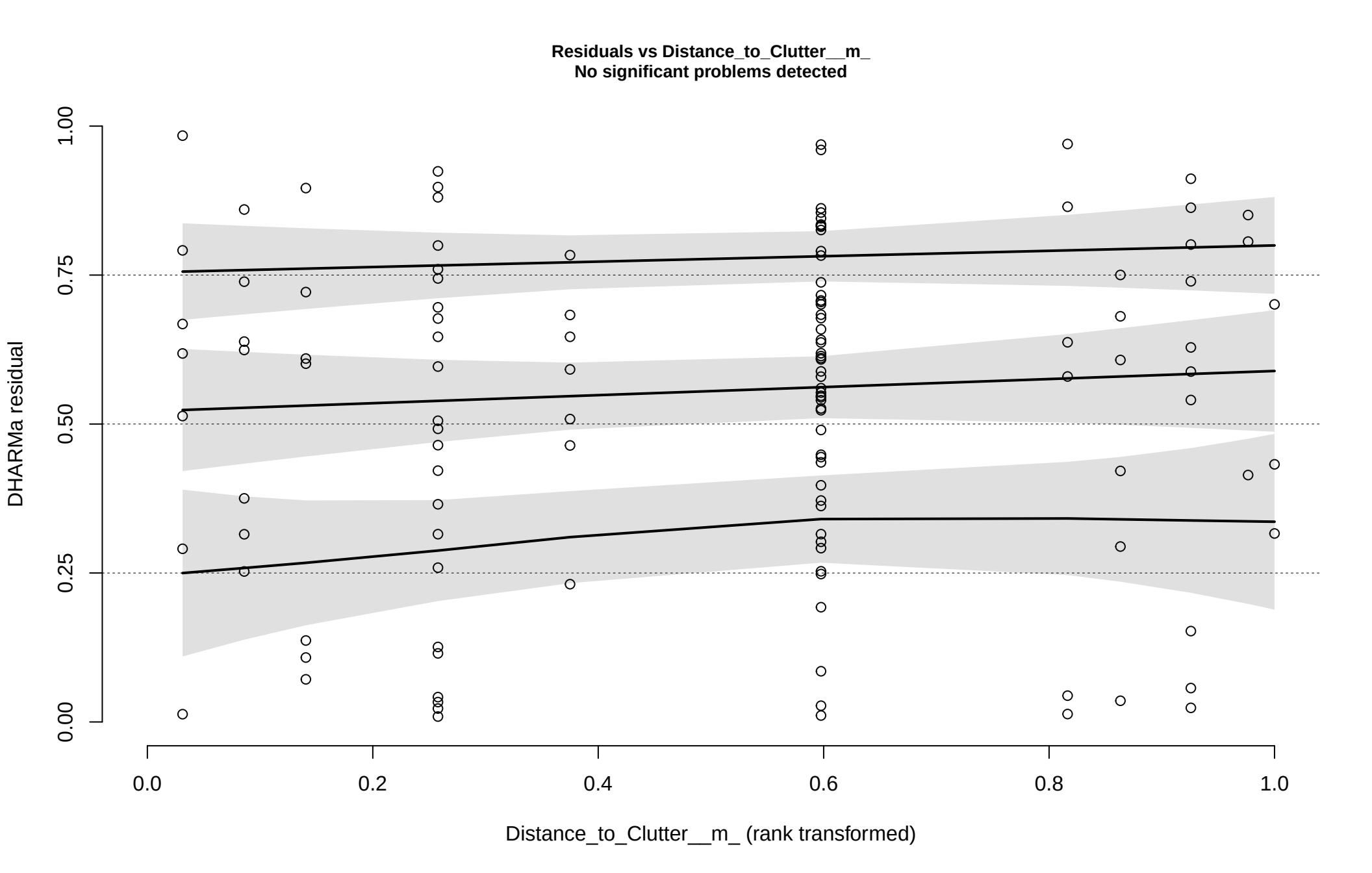


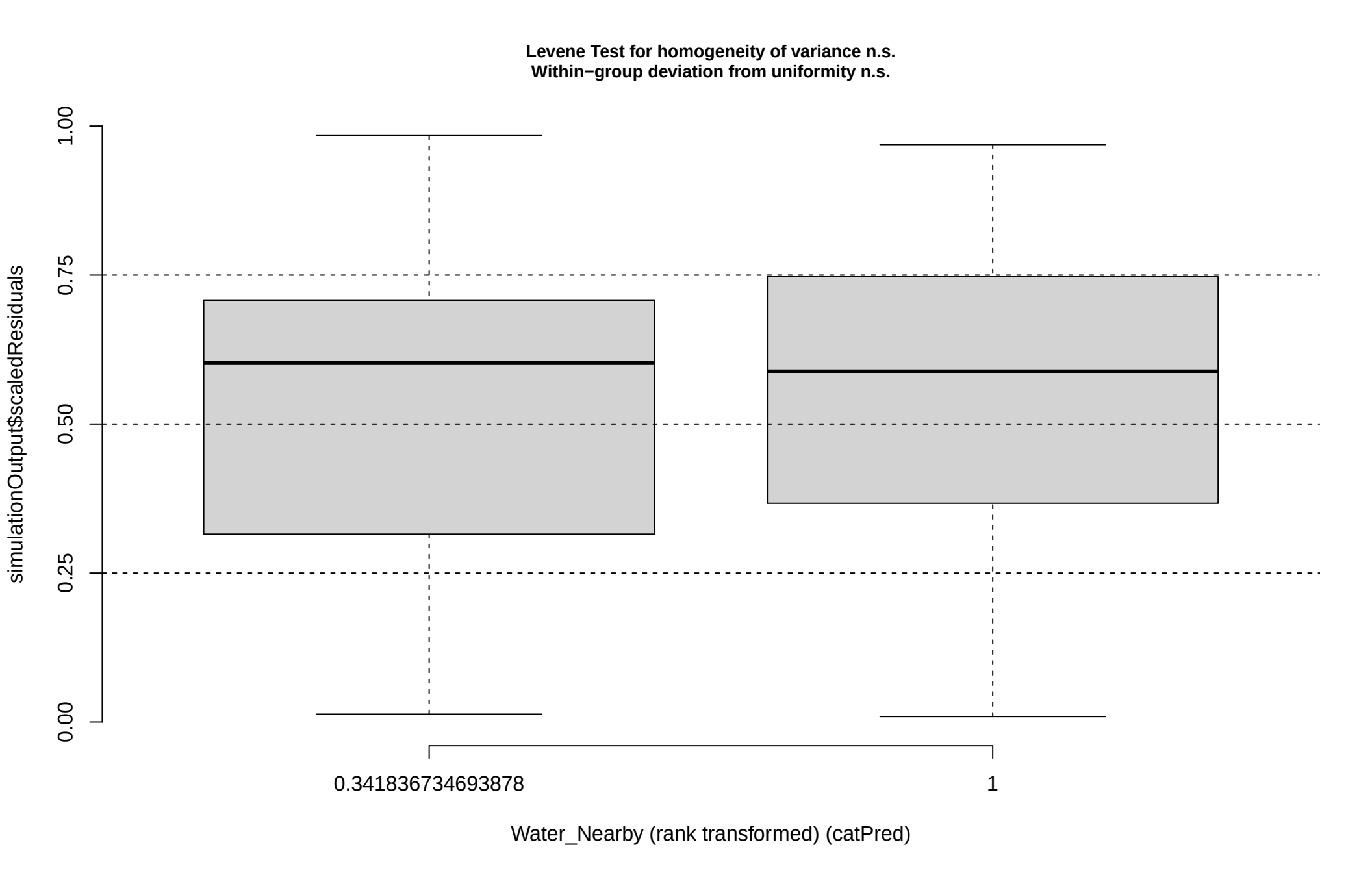


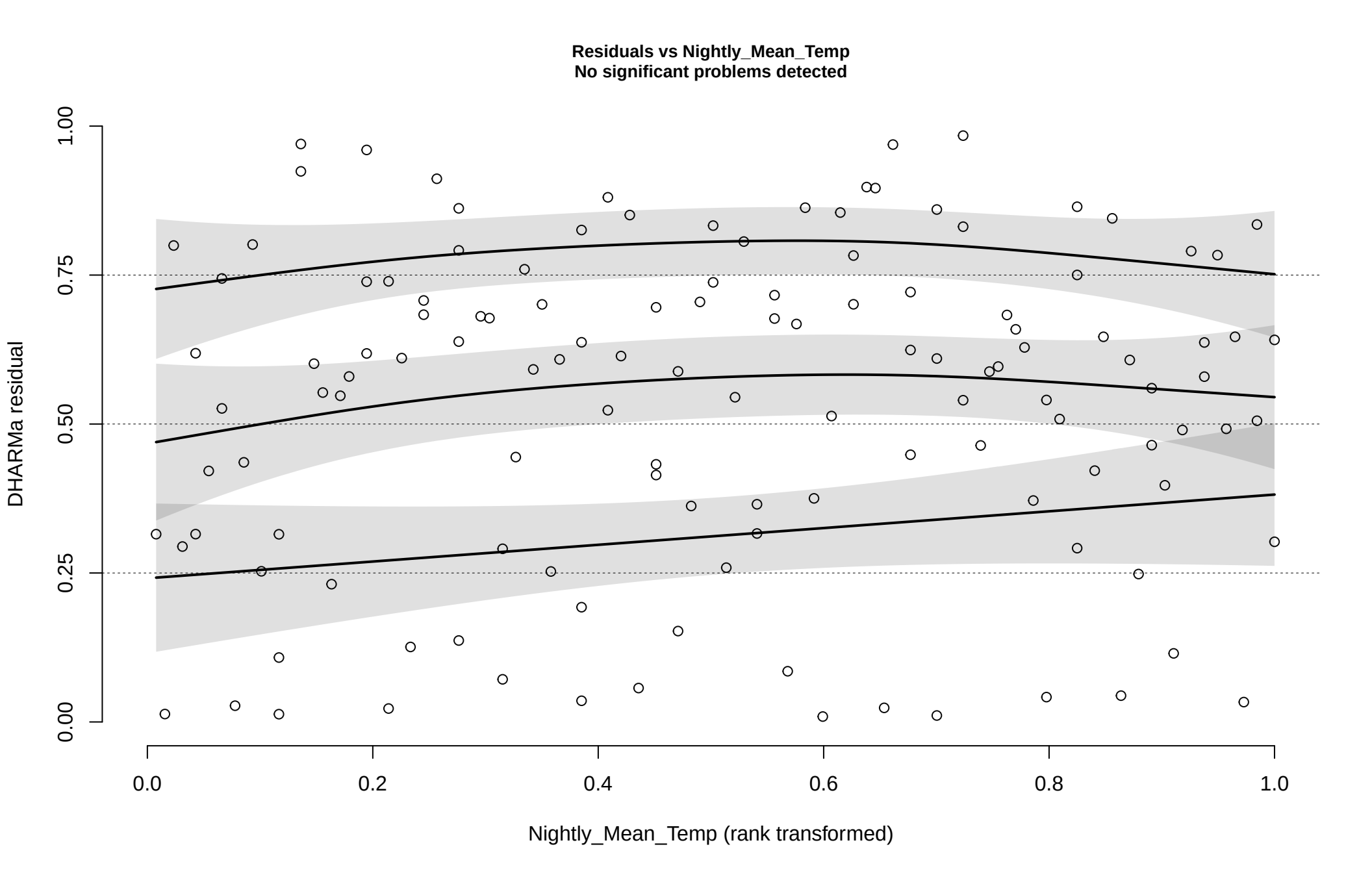

Residuals vs YearS
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

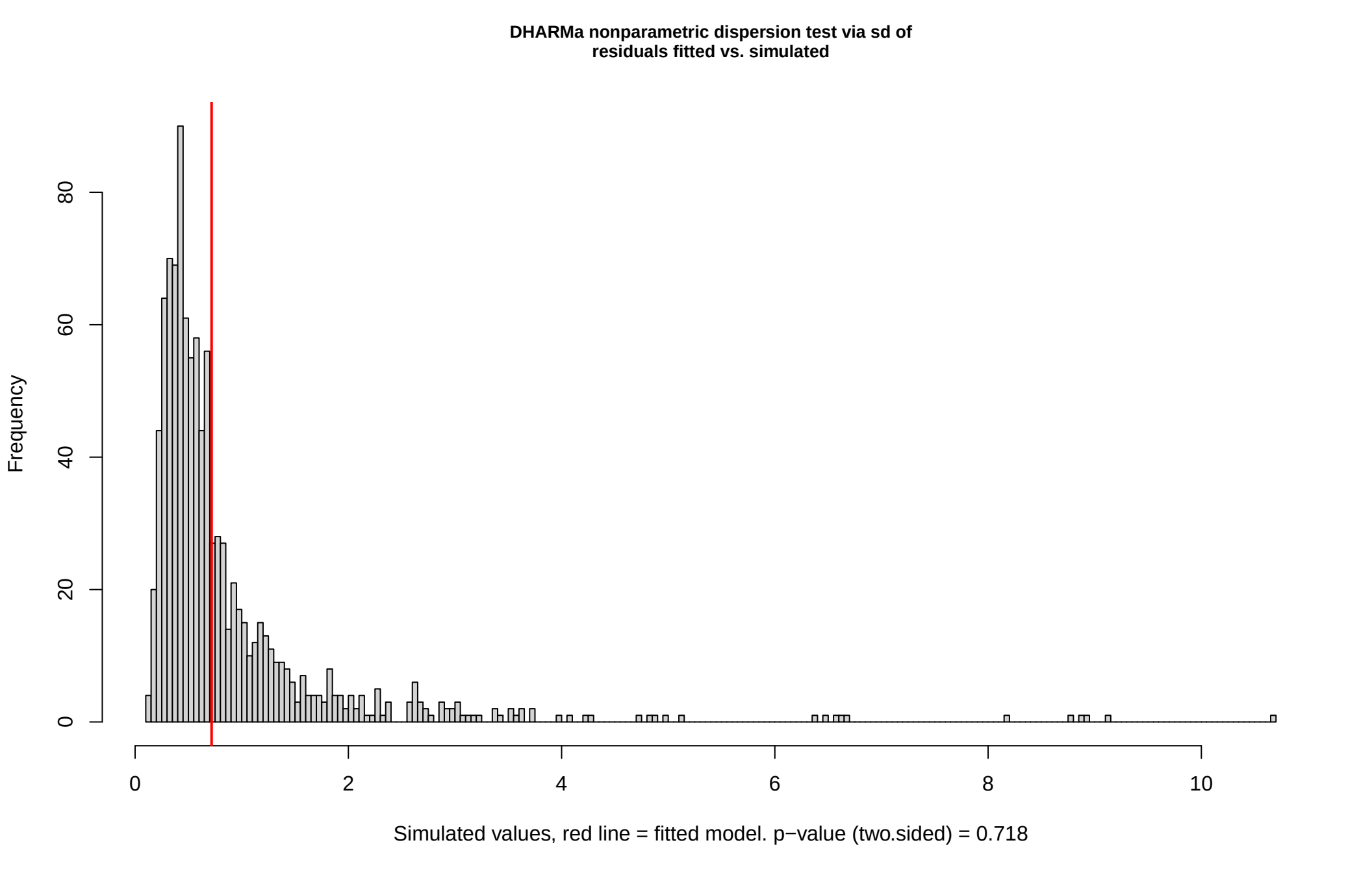

Residuals vs log(quad.nights)
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

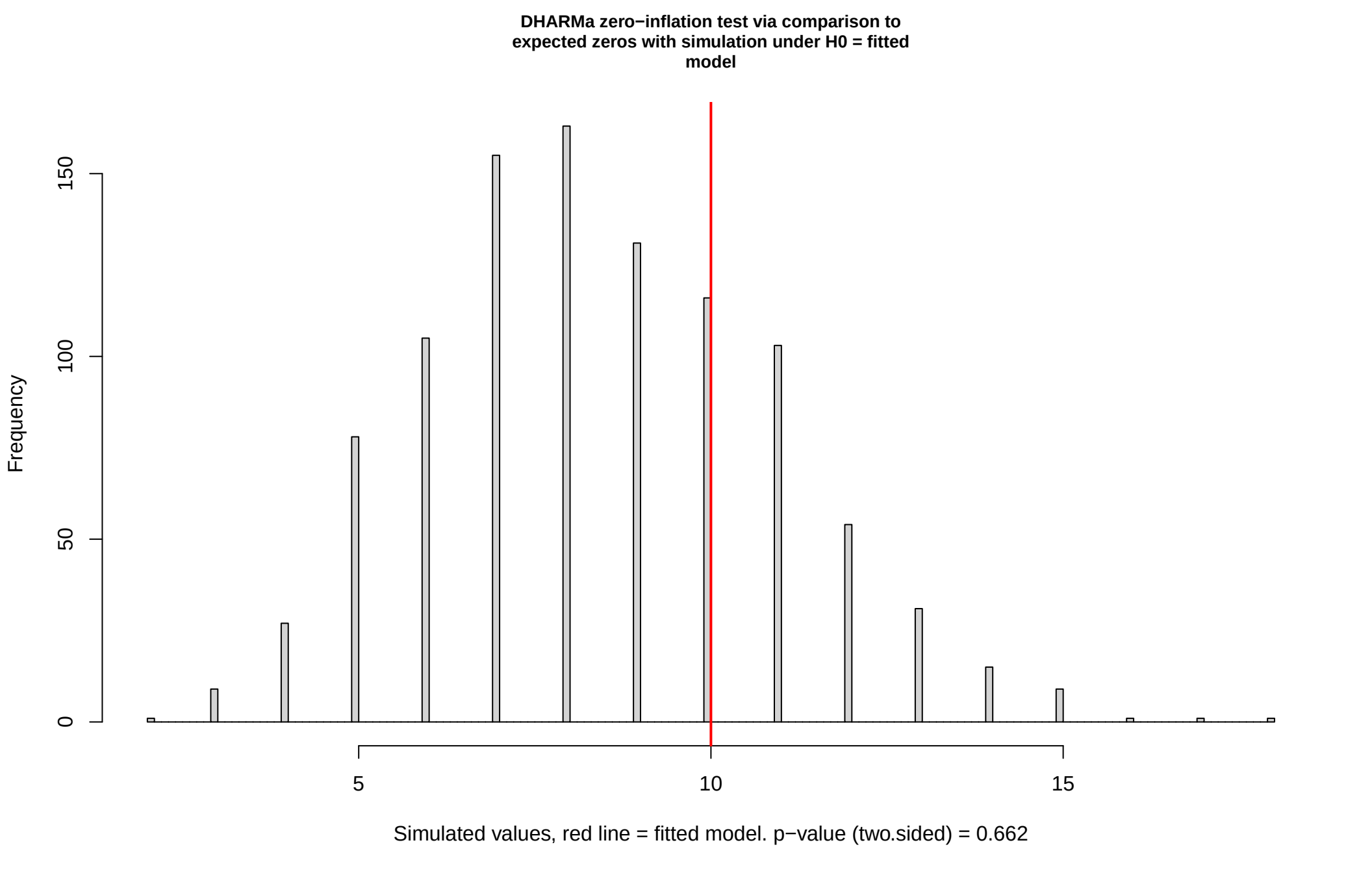

Residuals vs Nightly_Mean_Windsp
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Windsp (rank transformed)

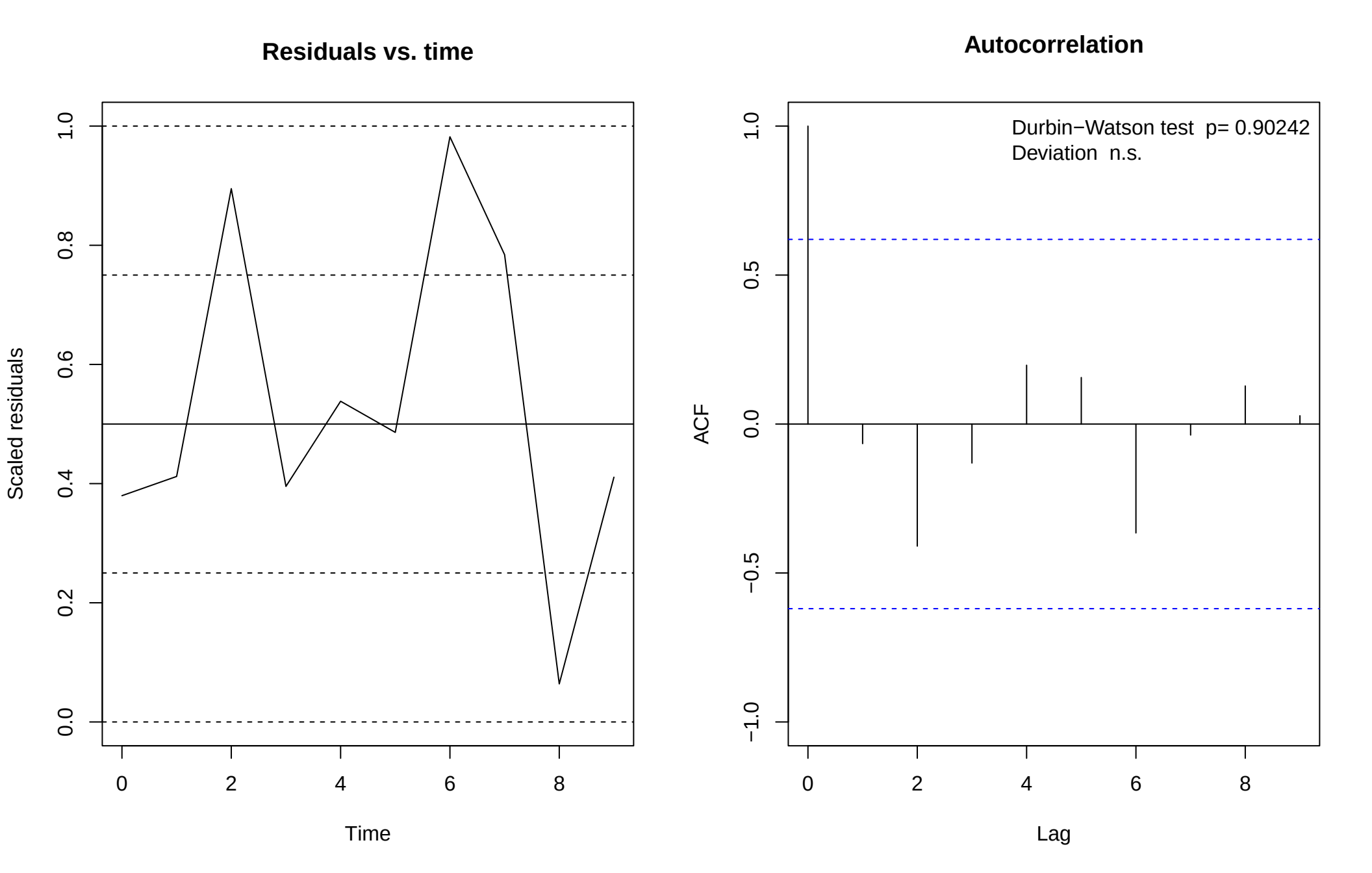

Residuals vs Distance_to_Clutter__m_
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Distance_to_Clutter__m_ (rank transformed)

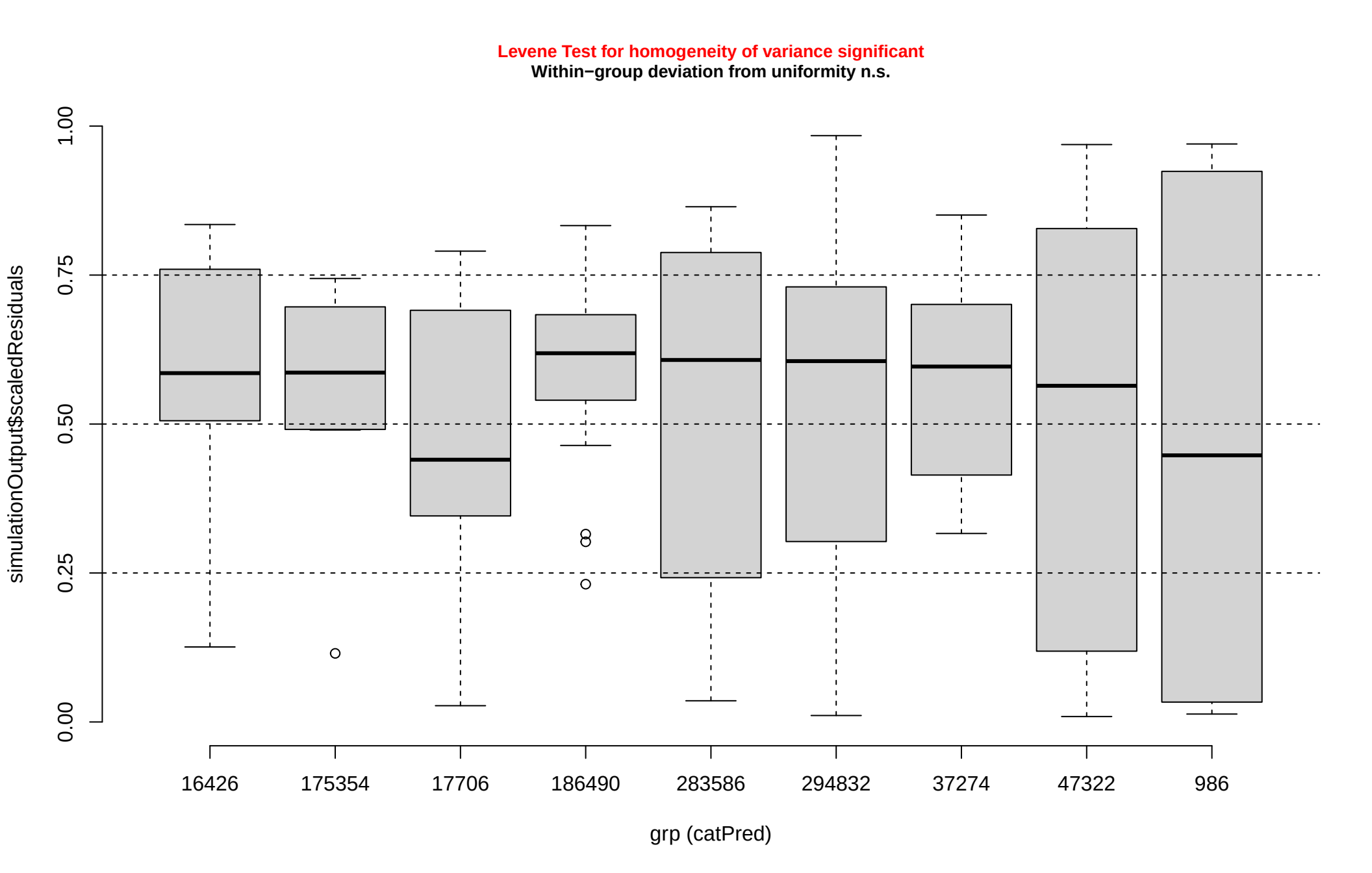

Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
0.341836734693878
1
Water_Nearby (rank transformed) (catPred)

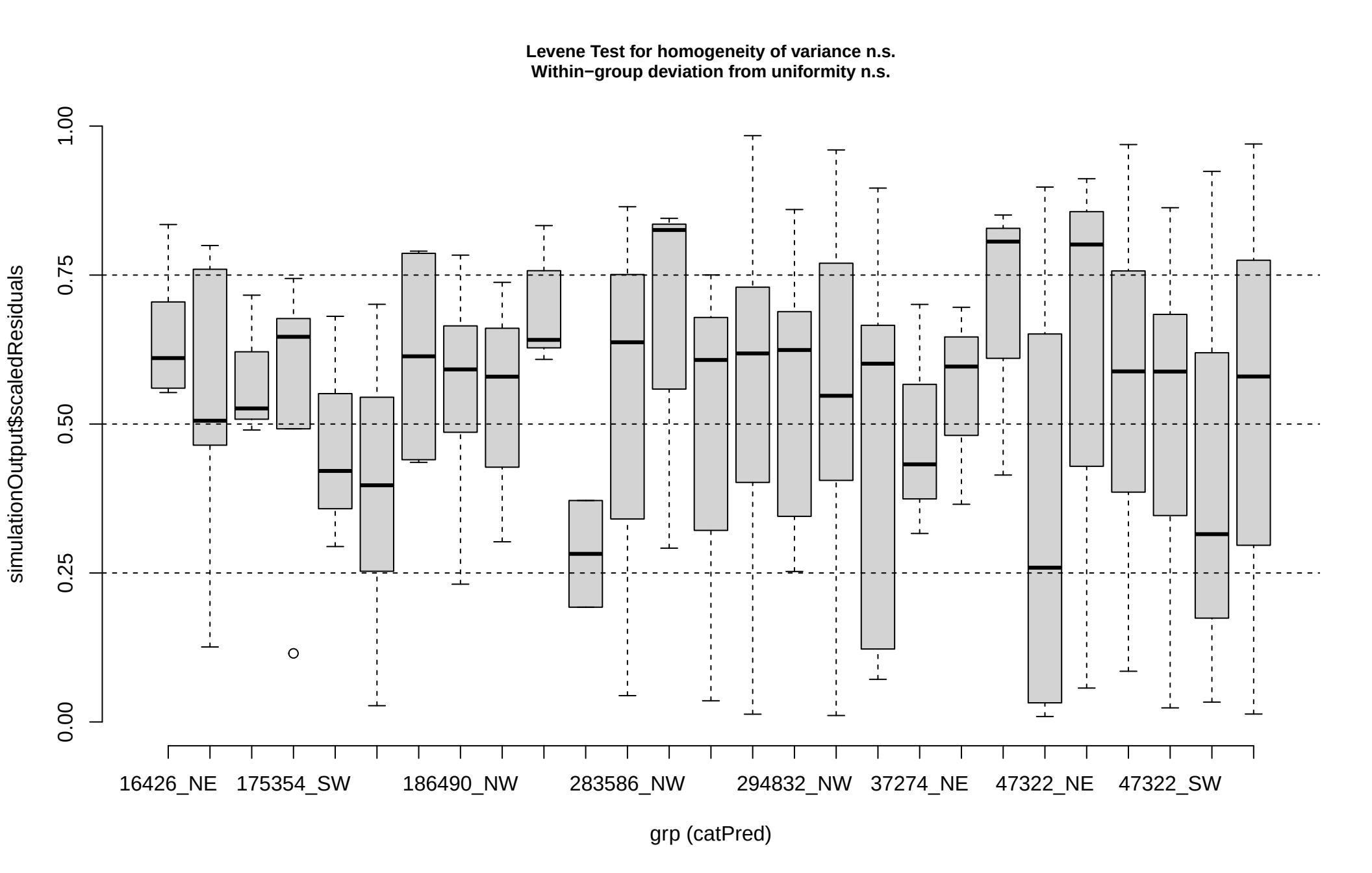

Residuals vs Nightly_Mean_Temp
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Temp (rank transformed)

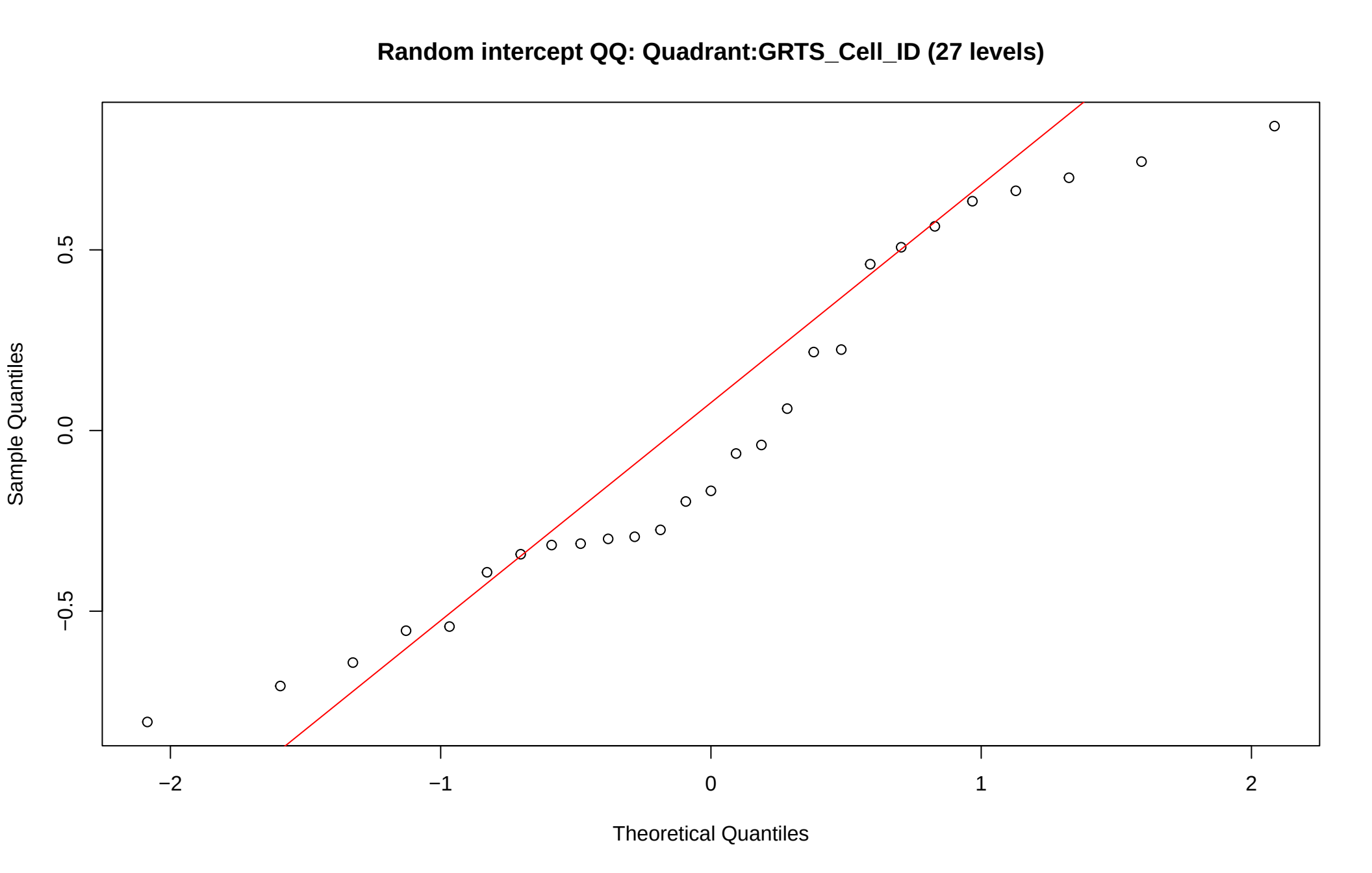

DHARMa nonparametric dispersion test via sd of
residuals fitted vs. simulated
Frequency
0
20
40
60
80
0
2
4
6
8
10
Simulated values, red line = fitted model. p−value (two.sided) = 0.718

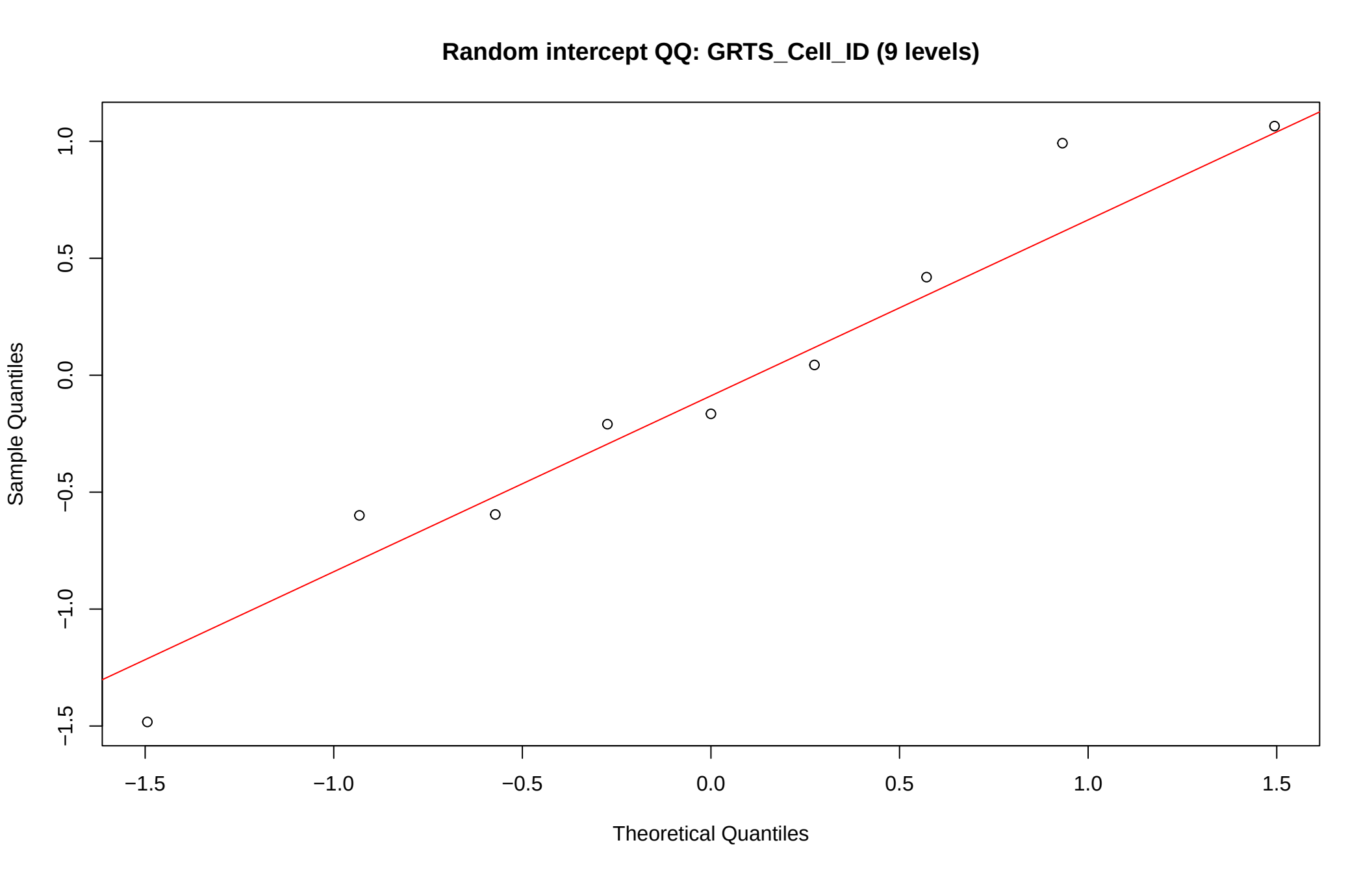

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
50
100
150
5
10
15
Simulated values, red line = fitted model. p−value (two.sided) = 0.662


Residuals vs. time
Scaled residuals
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
Time
Autocorrelation
ACF
−1.0
−0.5
0.0
0.5
1.0
Durbin−Watson test p= 0.90242
Deviation n.s.
0
2
4
6
8
Lag


Levene Test for homogeneity of variance significant
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
16426
175354
17706
186490
283586
294832
37274
47322
986
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
16426_NE
175354_SW
186490_NW
283586_NW
294832_NW
37274_NE
47322_NE
47322_SW
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (27 levels)
Sample Quantiles
0.5
0.0
−0.5
−2
−1
0
1
2
Theoretical Quantiles


Random intercept QQ: GRTS_Cell_ID (9 levels)
Sample Quantiles
Theoretical Quantiles
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5